

Principles of Inheritance and Variation

Complete Study Notes – PU-II Biology, Chapter 4

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Complete Study Notes – All 8 Topics

1. Genetics Basics & Mendel's Experiments

- **Genetics** is the branch of biology dealing with **inheritance** and **variation** of characters from parents to offspring.
- **Inheritance** – the process by which characters are passed on from parent to progeny.
- **Variation** – the degree by which the progeny differs from its parents.
- Humans knew as early as **8000–1000 B.C.** that a cause of variation was hidden in sexual reproduction, and exploited naturally occurring variations to selectively breed plants/animals – e.g. artificial selection gave Indian breeds like **Sahiwal cows** in Punjab.

Key terms used in genetics

- **Gene** – the functional unit of inheritance; contains information required to express a particular trait (Mendel's "factor").
- **Alleles** – genes which code for a pair of contrasting traits; slightly different forms of the same gene present at a specific locus on homologous chromosomes, determining alternate expression of the same character. Eg. Round & wrinkled, tall & short.
- **Locus** (pl. Loci) – the specific region/point on the chromosome where a gene or its alleles reside.
- **Phenotype** – the external, observable appearance of an organism (expressed in words, e.g. tall, dwarf).
- **Genotype** – the genetic constitution of an organism (expressed by symbols, e.g. TT, Tt, tt).
- **Homozygous** – paired genes are similar on homologous chromosomes for a character. Eg. TT, tt.
- **Heterozygous** – paired genes are dissimilar on homologous chromosomes for a character. Eg. Tt.
- **Hybrid** – the offspring of two parents differing in one or more characters. Eg. $TT \times tt \rightarrow Tt$.
- **Dominant gene** – expresses itself in the presence of a contrasting gene. **Recessive gene** – fails to express itself in the presence of a dominant gene.
- **Dominant character** – a character completely expressed even in the presence of a recessive allele.
- **Monohybrid** – a plant/animal obtained by crossing two homozygous individuals differing in one character.
- **Dihybrid** – a plant/animal obtained by crossing two homozygous individuals differing in two pairs of contrasting characters.
- **Back cross** – a cross between an F1 individual and either of its parents.
- **Test cross** – a backcross between an F1/F2 hybrid (unknown genotype) with a homozygous **recessive** parent.
- **Out cross** – a backcross between an F1/F2 hybrid (unknown genotype) with a homozygous **dominant** parent.

Mendel's experiment

- **Gregor Johann Mendel**, known as the **father of genetics**, proposed the laws of inheritance.
- Conducted hybridisation experiments on **garden pea** (*Pisum sativum*) for **seven years (1856–1863)**.
- First person to use **statistical analysis and mathematical logic** in biology; a large sampling size gave credibility to his data.
- He conducted artificial pollination / cross-pollination (hybridisation) using several **true breeding pea lines**.
- **True breeding pea lines** – lines that have undergone self-pollination for several generations and show stable inheritance and expression of traits.
- Mendel selected **14 true breeding pea lines as 7 contrasting pairs**, similar except for one character.

Character	Dominant trait	Recessive trait
Stem height	Tall	Dwarf
Seed shape	Round	Wrinkled
Seed colour	Yellow	Green
Pod shape	Inflated (full)	Constricted

Character	Dominant trait	Recessive trait
Pod colour	Green	Yellow
Flower colour	Violet	White
Flower position	Axial	Terminal

Why Mendel chose the pea plant

- Easy to cultivate; required minimum space and time.
- Possessed a number of easily distinguishable contrasting characters.
- Short life span.
- Flowers are naturally self-pollinating; cross-pollination was also possible by transferring pollen grains from one flower to another (artificial hybridisation, with emasculation).
- All pea plants produce a large number of seeds, giving strong statistical sample sizes.

2. Monohybrid Cross & Mendel's Laws

- Mendel performed artificial cross-pollination (hybridisation) on true breeding pea lines to obtain the first hybrid generation, the **first filial progeny (F1)**.
- After hybridisation, the F1 generation resembled only **one** of its parents (say, all tall; no dwarf).
- When F1 plants were self-pollinated (selfing), the **second filial progeny (F2)** was obtained.
- Revival of the unexpressed trait (dwarf) was observed in some F2 progeny – both tall and dwarf appeared in F2 in the ratio **3:1**.
- Mendel proposed that something is passed unchanged from generation to generation – he called these '**factors**' (presently, genes). Two alleles code for a pair of contrasting traits.

Monohybrid cross (single character, e.g. height)

- Parents: TT (tall) × tt (dwarf) → Gametes: T and t → F1: **Tt** (all heterozygous, tall).
- Selfing of F1 (Tt × Tt) → F2 genotypes **TT : Tt : tt = 1 : 2 : 1**; F2 phenotypes **Tall : Dwarf = 3 : 1**.
- TT, Tt, tt are **genotypes**; tall and dwarf are **phenotypes**. 'T' dominates and suppresses 't', so 'T' (tallness) is the **dominant trait** and 't' (dwarfness) is the **recessive trait**.
- TT and tt are **homozygous**; Tt is **heterozygous**. Alleles of a parental pair separate/segregate from each other, so only one allele is transmitted to a gamete: gametes of TT carry only T; gametes of tt carry only t; gametes of Tt carry both T and t (in equal proportion).

Mendel's Laws of Inheritance

- **Mendel's First Law / Law of Dominance** – characters are controlled by discrete units called factors, which occur in pairs, with one member of the pair dominating over the other in a dissimilar pair. Explains expression of only one parental character in F1, and expression of both in F2.
- **Mendel's Second Law / Law of Segregation (Law of Purity of Gametes)** – the two alleles of a pair segregate (separate) during gamete formation such that a gamete receives only one of the two factors. In homozygous parents, all gametes produced are similar; in heterozygous parents, two kinds of gametes are produced in equal proportions.
- **Mendel's Third Law / Law of Independent Assortment** – based on the dihybrid cross (covered next topic).
- Law of Dominance and Law of Segregation are based on the **monohybrid** cross; Law of Independent Assortment is based on the **dihybrid** cross.

Punnett square / Checker board

- A graphical method (developed by British geneticist **Reginald C. Punnett**) to calculate the probability of all possible genotypes of offspring in a genetic cross. Possible gametes are written along the top row and left column; combinations fill the boxes.

- From the Punnett square, F2 genotypic ratio $TT : Tt : tt = 1 : 2 : 1$ can be derived from the **binomial expression** $(ax + by)^2$, where 'a' and 'b' are the gamete-bearing genes T or t (each at frequency $\frac{1}{2}$), 'x' is the dominant allele and 'y' the recessive allele: $(\frac{1}{2}T + \frac{1}{2}t)^2 = \frac{1}{4}TT + \frac{1}{2}Tt + \frac{1}{4}tt$

Test cross

- A cross between an F1/F2 progeny and its homozygous recessive parent.
- Significance** – determines whether the dominant character is coming from a homozygous or heterozygous genotype (e.g. tallness coming from TT or Tt).
- Case 1:** TT (unknown, tall) $\times$ tt (homozygous recessive) $\rightarrow$ all progeny Tt (tall) $\rightarrow$ unknown plant is **homozygous dominant**.
- Case 2:** Tt (unknown, tall) $\times$ tt (homozygous recessive) $\rightarrow$ progeny 1 Tt (tall) : 1 tt (dwarf) $\rightarrow$ unknown plant is **heterozygous**.
- In a **dihybrid** test cross, the ratio obtained is **1:1:1:1**.

Monohybrid cross in animals – coat colour in guinea pig

- In guinea pigs (*Cavia porcellus*), black coat (BB) is dominant over white coat (bb).
- Homozygous black male $\times$ homozygous white female $\rightarrow$ F1 all black, heterozygous (Bb) – each sperm carries B, each egg carries b.
- Selfing F1 (Bb $\times$ Bb) $\rightarrow$ F2 phenotypic ratio Black : White = 3 : 1; genotypic ratio BB : Bb : bb = 1 : 2 : 1.

3. Dihybrid Cross & Independent Assortment

- A cross between two homozygous individuals differing in **two pairs** of contrasting characters is called a **dihybrid cross** (e.g. seed shape and seed colour).
- Round shape and yellow colour are dominant over wrinkled shape and green colour.
- Homozygous pea plant producing round, yellow seeds (**RRYY**) $\times$ homozygous plant producing wrinkled, green seeds (**rryy**) – these are the Parent generation (P).
- F1 offspring: round, yellow seeds (**RrYy**) – all heterozygous.

F2 generation (selfing RrYy $\times$ RrYy)

- Each parent produces **4 types of gametes** (RY, Ry, rY, ry), each at a frequency of 25%.
- Punnett square produces **16 combinations**, giving **16 types** of F2 plants: round yellow, round green, wrinkled yellow, and wrinkled green in the ratio **9 : 3 : 3 : 1**.
- Of the F2 plants, one is RRYYY (pure dominant) and one is rryy (pure recessive, same as parental types). The other **14 types are new combinations** arising from **recombination of genes**.
- Phenotypic ratio** – 9 : 3 : 3 : 1, i.e. 9 Round Yellow : 3 Round Green : 3 Wrinkled Yellow : 1 Wrinkled Green.
- Genotypic ratio** – 1 : 2 : 1 : 2 : 4 : 2 : 1 : 2 : 1 (i.e. 1 RRYYY : 2 RRYy : 1 RRyy : 2 RrYY : 4 RrYy : 2 Rryy : 1 rrYY : 2 rrYy : 1 rryy).
- The 9:3:3:1 ratio can be derived as a combination series of **(3 Round : 1 Wrinkled) $\times$ (3 Yellow : 1 Green) = 9 Round,Yellow : 3 Round,Green : 3 Wrinkled,Yellow : 1 Wrinkled,Green**.

Mendel's Third Law / Law of Independent Assortment

- When two pairs of traits are combined in a hybrid, one pair of traits segregates **independently** of the other pair.
- When each pair of contrasting characters is considered singly, they separate independently – segregation of one trait's contrasting characters is totally independent of segregation of another trait's characters.
- Genes for shape do not intermix with genes for colour, but **assort independently**.
- In the dihybrid cross between RRYYY and rryy, four types of gametes (RY, Ry, rY, ry) are produced, each segregating independently at a frequency of 25% of the total gametes.

Rediscovery of Mendel's work

- Mendel published his work in the journal '**Proceedings of the Natural History Society**' of Brunn in **1865**, but it remained unrecognised till 1900 because: Lack of communication and publicity.
- His concept of factors (genes) as **discrete units** that did not blend with each other was not accepted, in light of variations occurring continuously in nature.
- Mendel's approach of explaining a biological phenomenon with the help of **mathematics** was also not accepted at the time.

4. Deviations from Mendelism

Incomplete Dominance / Intermediate Inheritance

- The F1 generation has a phenotype that does **not resemble either** of the two parents, but is a **mixture** of the two.
- Discovered by **Carl Correns** in 1903 in the Four O'clock plant.
- Example 1 – *Antirrhinum majus*** (snapdragon/dog flower) & ***Mirabilis jalapa*** (Four O'clock plant): RR (red) × rr (white) → F1 all Rr (**pink**). Selfing F1 → F2 Red : Pink : White = **1 : 2 : 1** (phenotypic ratio same as genotypic ratio, since dominance is not complete).
- Example 2 – Andalusian fowls:** Black (BB) × White (bb) → F1 all **blue** (Bb) due to fine alternate black & white stripes on feathers (favoured as a delicacy). F2 – 1 Black : 2 Blue : 1 White.

Dominance	Incomplete Dominance
F1 is similar to the dominant parent	F1 is different from either of the two parents
Phenotypic ratio differs from genotypic ratio	Phenotypic and genotypic ratios are the same
In F1 hybrid, dominant trait is completely expressed	In F1 hybrid, dominant trait is incompletely expressed
A single dominant allele (e.g. Tt) expresses fully	Two alleles (e.g. RR) are required for full expression of the dominant trait

Explanation of the concept of dominance

- A diploid organism has two copies of a gene (a pair of alleles); the two alleles need not be identical (heterozygote), as one may be modified.
- Suppose a gene produces an enzyme that transforms substrate S. The modified allele could produce: (i) the normal/less efficient enzyme (an **equivalent allele**, same phenotype), (ii) a non-functional enzyme, or (iii) no enzyme at all.
- The original gene is **dominant**; the modified gene is **recessive**. The recessive trait is seen due to a non-functional enzyme or because no enzyme is produced – the phenotype then depends only on the unmodified (dominant) allele's functioning.

Multiple Allelism / Gene Polymorphism

- Expression of a single gene in **more than two** alternative forms is called **multiple allelism** or gene polymorphism – found only when population studies are made.
- Example – ABO blood groups:** controlled by gene "I", which has three alleles – **I^A, I^B, and i**. I^A and I^B dominate over i, but with each other, I^A and I^B are **co-dominant**. I^A and I^B produce A and B type sugars (glycoprotein/antigens), while i produces no sugar.
- ABO blood groups discovered by **Karl Landsteiner** (1901); the inheritance pattern was worked out by **Felix Bernstein** (1924).

Allele from Parent 1	Allele from Parent 2	Genotype of offspring	Blood group
I ^A	I ^A	I ^A I ^A	A
I ^A	I ^B	I ^A I ^B	AB
I ^A	i	I ^A i	A
I ^B	I ^B	I ^B I ^B	B
I ^B	i	I ^B i	B
i	i	ii	O

- There are **6 genotypes** ($I^A I^A$, $I^A i$, $I^B I^B$, $I^B i$, $I^A I^B$, ii) but only **4 phenotypes** (A, B, AB, O) for the ABO blood group.
- Homozygous "A" parents produce only "A" progeny; heterozygous "A" parents produce "A" and "O" progeny (similarly for "B"); "AB" parents produce "A", "B" & "AB" progeny; "O" parents produce only "O" progeny.

Co-dominance

- The phenomenon of expression of **both** alleles in a heterozygote is called **co-dominance**. Co-dominant alleles do not show a dominant-recessive relationship and express independently when present together; the F1 progeny resembles **both** parents.
- **Example 1 – ABO blood group:** "AB" blood group – I^A and I^B are co-dominant.
- **Example 2 – Sickle cell anaemia:** a hereditary disease; the sickle-cell gene Hb^S is co-dominant with the normal allele Hb^A . Genotype $Hb^A Hb^A$ = normal haemoglobin; $Hb^S Hb^S$ = sickle-cell haemoglobin (RBCs sickle under low O_2 tension, patients die of lethal/fatal anaemia); heterozygous $Hb^A Hb^S$ individuals have both types of haemoglobin and suffer only mild anaemia (some RBCs sickle during O_2 deficiency) – showing Hb^A and Hb^S are co-dominant.
- **Example 3 – Coat colour in cattle:** gene R = red coat, W = white coat. Red (RR) × White (WW) → F1 hybrids are **roan** coloured (genotype RW) – dark (red) colour interspersed with white, developing side by side in the heterozygote. F2 – red, roan and white cattle in the ratio **1 : 2 : 1** (RR : RW : WW).

Incomplete Dominance	Co-Dominance
Effect of one of the two alleles is more conspicuous	Effect of both alleles is equally conspicuous
Produces a fine mixture of the expression of the two alleles	No mixing of the effect of the two alleles
Effect in hybrid is intermediate of the expression of two alleles	Both alleles produce their effect independently, e.g. I^A & I^B , Hb^S & Hb^A
Hybrid possesses a new phenotype	New phenotype does not develop
Expressed phenotype has no allele of its own	Expressed phenotype is a combination of two phenotypes and their alleles

Pleiotropy

- A condition in which a **single gene** produces **more than one trait** (multiple phenotypic expression) is called **pleiotropy**; such a gene is a **pleiotropic gene**. It mainly affects metabolic pathways that contribute towards various phenotypic characters.
- **Example 1 – Starch synthesis in pea seeds:** controlled by one gene with two alleles "B" and "b". "BB" homozygotes synthesise starch effectively, producing **large starch grains** (round seeds); "bb" homozygotes are less efficient, producing **smaller starch grains** (wrinkled seeds); heterozygotes "Bb" produce **round seeds** (so B seems dominant), but with starch grains of **intermediate size**. So from the angle of starch-grain size (the true phenotype), the alleles show incomplete dominance – when a gene produces more than one phenotype, dominance is a feature of the gene product examined, not the gene itself.
- **Example 2 – Phenylketonuria (PKU):** discovered by **Folling (1934)**; caused by a single-gene mutation (on chromosome 12) in the gene coding for the enzyme **phenylalanine hydroxylase**, which converts phenylalanine to tyrosine (needed for protein synthesis and formation of catecholamine, thyroxine and melanin). Absence of the enzyme leads to conversion of phenylalanine to phenylpyruvic acid, which accumulates in the brain (causing mental retardation), reduces hair/skin pigmentation, and is excreted through urine (poor kidney absorption). Occurs in about 1 in 18000 births among white Europeans; very rare in other races; blood/urine tests of neonates can detect it.
- **Example 3 – Sickle cell anaemia:** the causative gene alters both the **type of haemoglobin** and the **form/shape of the RBC**.

Polygenic Inheritance / Multifactorial or Quantitative Inheritance

- A trait controlled by **three or more genes** is a **polygenic trait**; such genes are called **polygenes / multiple genes / cumulative genes**.

- The effect of each allele is **additive/cumulative**; besides multiple genes, environment also influences the phenotype.
- **Example – Human skin colour**: assume three genes A, B, C control skin colour; the dominant form "ABC" gives dark skin, recessive "abc" gives light skin. AABBCC has the **darkest** skin colour, aabbcc the **lightest**; AaBbCc (three dominant + three recessive alleles) gives an **intermediate** skin colour. The number of dominant alleles present determines the darkness/lightness of skin.
- **F. Galton** studied polygenic inheritance in man and coined the term **eugenics** (application of genetics to attempt to improve the hereditary qualities of humans).
- **Nilsson-Ehle** obtained the first experimental proof of polygenic inheritance, in the case of kernel colour of wheat.

5. Chromosomal Theory of Inheritance & Linkage

Chromosomal Theory of Inheritance

- By 1900, advancement in microscopy allowed chromosomes to be discovered.
- **Walter Sutton** and **Theodore Boveri** discovered that the behaviour of chromosomes was parallel to the behaviour of genes.
- Chromosomes and genes both occur in **pairs** – two alleles of a gene pair are located at **homologous sites on homologous chromosomes**.
- Sutton and Boveri proposed that it is the **pairing and separation** of a pair of chromosomes that ultimately leads to **segregation** of the pair of factors (genes) they carry.
- During **Anaphase I of meiosis**, the two chromosome pairs that were aligned at the metaphase plate separate out **randomly** – this independent alignment/separation is the physical basis of independent assortment.
- Union of the knowledge of chromosomal segregation with Mendelian principles constitutes the **chromosomal theory of inheritance**.

Genes	Chromosomes
Occur in pairs	Occur in pairs
Segregate at gamete formation so only one of each pair is transmitted to a gamete	Segregate at gamete formation so only one of each pair is transmitted to a gamete
Independent pairs segregate independently of each other	One pair segregates independently of another pair

Linkage and Recombination – Morgan's experiment

- **Thomas Hunt Morgan** ('father of experimental genetics' / 'father of sex linkage') discovered the basis of variations produced by sexual reproduction.
- He worked on **fruit flies** (*Drosophila melanogaster*), chosen because: they grow on synthetic medium in the laboratory; their life cycle completes in **two weeks**; a single mating produces many progeny flies; clear differentiation of sexes (easily distinguishable male/female); hereditary variations are clearly visible with low power microscopes.
- **Morgan's experiment**: a dihybrid cross was carried out – yellow-bodied, white-eyed females crossed with brown-bodied, red-eyed males; F1 progeny obtained and inter-crossed; F2 progeny obtained and its ratio observed to be **significantly different** from the 9:3:3:1 ratio seen in a typical Mendelian dihybrid cross.

Explanation of the deviation

- The genes involved were located on the **same (X) chromosome**. When two genes are located on the same chromosome, the proportion of **parental gene combinations** was much higher than that of non-parental (new) combinations.
- **Linkage** – physical association of genes on a chromosome.
- **Recombination** – generation of non-parental gene combinations.
- **Cross A** (genes y & w, body colour & eye colour): Parental type = **98.7%**, Recombinant type = **1.3%** – genes y and w are **tightly linked**, showing low recombination.

- **Cross B** (genes w & m, eye colour & wing size): Parental type = **62.8%**, Recombinant type = **37.2%** – genes w and m are **loosely linked**, showing high recombination.
- The strength of linkage between y and w is **higher** than between w and m.

Low Recombination	High Recombination
Occurs when genes on the same chromosome are tightly linked	Occurs when genes on the same chromosome are loosely linked
Example – genes white & yellow are tightly linked, show only 1.3% recombination	Example – genes white & miniature are loosely linked, show 37.2% recombination

Genetic Mapping

- **Alfred Sturtevant** utilised the frequency of gene recombination as a measure of the physical distance between two genes, to map their position on chromosomes. Genetic maps prepared this way are extensively used today for genome sequencing projects, such as the Human Genome Project.
- Distance between genes is measured by the **map unit (m.u.)**. **1% crossing over** between two linked genes = **1 map unit** or **Centi Morgan (cM)**; 10% crossing over = **deci Morgan (dM)**; 100% crossing over = **Morgan (M)**.
- The physical distance between two genes determines both the strength of linkage and the frequency of crossing over. The strength of linkage **increases** with the **closeness** of the two genes; the frequency of crossing over **increases** with the **increase** in physical distance between the two genes.

6. Sex Determination

- **Henking** discovered the genetic/chromosomal basis of sex determination while working on insects. He observed specific nuclear structures during spermatogenesis, naming them **X bodies**; he found that after spermatogenesis, only 50% of sperm obtained these structures, while 50% did not.
- The X body was later found to be a chromosome, and was named the **X chromosome**.
- Chromosomes involved in sex determination are called **allosomes (sex chromosomes)**; the others are called **autosomes**.

XO type of sex determination

- Besides autosomes, at least **one X chromosome** is present in all insects.
- Some sperm contain an X chromosome, while some do not (no Y chromosome exists in this system).
- Eggs fertilised by X-bearing sperm become **females (XX)**; eggs fertilised by sperm lacking X become **males (XO)**.
- Example of organisms with XO type sex determination – **Grasshopper, Earthworm**.

XY type of sex determination

- The chromosome pattern in the human female is **XX** and in the male is **XY**.
- All haploid gametes produced by the female (ova) carry the sex chromosome X. Males produce 50% of sperm carrying the X chromosome, while the other 50% carry Y.
- After fertilisation, the zygote carries either XX or XY depending on whether the X-bearing or Y-bearing sperm fertilised the ovum. The zygote carrying XX develops into a **female**; XY develops into a **male**.
- Example of organisms with XY type sex determination – **Humans and Drosophila**.

Male heterogamety and Female heterogamety

- **Male heterogamety** – XO and XY types of sex determination are examples. In XO type, some gametes have an X chromosome, while some are without X. In XY type, some gametes have X chromosomes, while some have Y chromosomes.
- **Female heterogamety** – the **ZW type** of sex determination is an example. Here, the female has one Z and one W chromosome, while the male has a pair of Z chromosomes. Example – **Birds and some reptiles and fishes**.

Haplodiploid Sex-Determination in Honey Bee

- Sex determination in honey bees is based on the **number of sets of chromosomes** an individual receives.

- An offspring formed from the union of a sperm and an egg develops as a **female** (queen or worker); an **unfertilised** egg develops as a **male** (drone) by **parthenogenesis**.
- This means males have **half** the number of chromosomes of a female. Females are **diploid**, having **32 chromosomes**; males are **haploid**, having **16 chromosomes**. This is called the **Haplodiploid sex-determination system**.
- Males produce sperm by **mitosis** – they do not have a father and thus cannot have sons, but have a grandfather and can have grandsons.

7. Mutation & Pedigree Analysis

Mutation

- **Mutation** – alteration of the DNA sequence, resulting in changes in the genotype and phenotype of organisms.
- **Mutagens** – substances which cause mutation: **Physical mutagens** – UV radiation, ionizing radiation, gamma rays.
- **Chemical mutagens** – nitrous acid (a deaminating agent which changes cytosine to uracil).
- **Biological mutagens** – viruses (e.g. HPV), bacteria (e.g. *Helicobacter*).

Types of mutation

- **Point Mutation** – a mutation arising due to change in a **single base pair** of DNA, as in sickle cell anaemia.
- **Frameshift Mutation** – a mutation arising due to **deletion or insertion** of a base pair in the DNA sequence.

Pedigree Analysis

- **Pedigree analysis** is the analysis of inheritance of a trait across several generations of a family; the trait under study is represented in a **family tree**.
- Using pedigree analysis, the inheritance of a specific trait, abnormality, or disease can be traced.
- DNA is believed to be the carrier of genetic information, which passes unaltered from generation to generation. Mutations occasionally alter the genetic material, and genetic diseases are believed to be associated with these alterations only.

Standard symbols in pedigree analysis

- **Square** – Male; **Circle** – Female; **Diamond** – Sex unspecified.
- **Filled (shaded) shapes** – Affected individuals.
- A horizontal line joining a square and circle – **Mating**; a double horizontal line – **Mating between relatives (consanguineous mating)**.
- Parents are placed above and children below, in order of birth from left to right.
- A diagonal line through a symbol – **Deceased (death)**; an arrow pointing to an individual – the **Propositus/Proposita** (the person serving as the starting point of the genetic study of a family).
- A number inside a diamond (e.g. "5") – that many **unaffected offspring**.
- A small filled circle on a vertical line – **Abortion or still birth** (sex unspecified).
- A dot inside a circle/square, or a half-filled shape – **Carrier** of the trait (heterozygous, unaffected).
- Two circles/squares joined at a point below the parents – **Dizygotic twins**; joined at a point that itself joins a single line upward – **Identical (monozygotic) twins**.

8. Genetic Disorders

- **Genetic disorders** include **Mendelian disorders** and **Chromosomal disorders**.

Mendelian Disorders

- Characterised by **mutation in a single gene**; their mode of inheritance follows the principles of Mendelian genetics.

- Mendelian disorders can be: **(1) Autosomal dominant trait, (2) Autosomal recessive trait, (3) Sex-linked trait.**

1. Autosomal dominant trait

- The encoding gene is present on any of the autosomes, and the **wild type allele is recessive** to its mutant allele, i.e. the **mutant allele is dominant**.
- Examples – Muscular/Myotonic dystrophy, Huntington disease (HD) (chromosome 4), Brachydactyly, Polydactyly, Dimple in the cheek, Widow's peak, Tongue rolling, Tasting PTC (phenylthiocarbamide/phenylthiourea).
- **Characteristic features of inheritance:** transmission of traits occurs from parents of either sex; males and females are equally affected; the trait is present in each generation (**vertical pedigree**); multiple generations are characteristically affected.

2. Autosomal recessive trait

- The **mutant allele is recessive** to its wild type allele.
- Examples – Sickle cell anaemia (chromosome 11), Thalassaemia (chromosome 16 and 11), Phenylketonuria (chromosome 12), Cystic fibrosis (chromosome 7), Albinism, Fused ear lobe, Alkaptonuria (black urine disease), Gaucher's disease (chromosome 1), Tay-Sachs disease (TSD) / Infantile amaurotic idiocy / GM2 gangliosidosis.
- **Characteristic features of inheritance:** occur in equal proportion in multiple male and female siblings, whose parents are normal but carriers. The siblings are homozygous for the defective allele, but their parents (though appearing normal) are heterozygous carriers of the trait. Consanguinity (marriage between genetically related man and woman, e.g. cousins) occasionally results in the appearance of such traits.

Sickle cell anaemia

- An **autosomal recessive** disease, transmitted from parent to offspring when both parents are carriers.
- The pair of alleles Hb^A and Hb^S controls the expression: Hb^AHb^A – normal; Hb^AHb^S – carrier of disease; Hb^SHb^S – diseased.
- **Cause of the disease** – a change in the gene at **chromosome no. 11** causes the replacement of GAG by GUG, leading to the substitution of Glu by Val at the **6th position of the beta globin chain** of haemoglobin. The mutant haemoglobin so formed polymerises at low oxygen tension, resulting in a change of shape of the RBC to **sickle-like**.

Thalassaemia

- Gr. Thalassa = Sea (i.e. Mediterranean); Anaemia = weak blood. An **autosomal linked recessive** blood disease.
- Transmission – from parents to offspring when both partners are unaffected carriers for the gene (heterozygous).
- The body makes an abnormal, **quantitatively** deficient form of haemoglobin, resulting in excessive destruction of RBC, which leads to anaemia.
- Haemoglobin is made of two polypeptide chains: **Alpha globin chain** and **Beta globin chain**. Thalassaemia occurs when there is a defective gene(s) for any one of the chains.
- **Alpha thalassaemia** – production of the alpha globin chain is affected; alpha globin chains are encoded by two closely linked genes (**HBA1 & HBA2**) on **chromosome 16**; occurs in people from Southeast Asia, the Middle East and China.
- **Beta thalassaemia** – production of the beta globin chain is affected; beta globin chains are encoded by a single gene (**HBB**) on **chromosome 11** of each parent; occurs due to mutation in one or both of these genes.

3. Sex-linked trait

- **X-linked dominant trait** – Duchenne muscular dystrophy.
- **X-linked recessive trait** – Haemophilia, Colour blindness.
- **Y-linked trait** – Hypertrichosis.

X-linked dominant trait

- The encoding gene is present on the X chromosome, and the **mutant allele is dominant** over its wild type allele.
- **Example – Duchenne muscular dystrophy.** The dominant mutant allele is denoted 'D', and its recessive wild type allele 'd'. Human females have two X chromosomes (XX), and males have one X and one Y. Males receive

their lone X chromosome from their mother and the Y chromosome from their father, whereas females receive one X chromosome from their mother and the other X from their father.

- **Characteristics of inheritance:** the trait appears in almost all generations (vertical inheritance). If the female is affected, then about half of her sons and daughters are affected. If the male is affected, all of his daughters would be affected, but **none** of his sons are affected. In short, the pedigree resembles the pattern of autosomal dominant inheritance, except there is **no male-to-male transmission**.

X-linked recessive trait

- The gene is present on the X chromosome, and its **mutant allele is recessive** to its wild-type allele.
- Examples – **Haemophilia, Red-green colour blindness (Daltonism).**
- **Characteristic features:** females express the trait only when they are homozygous for the mutant allele, whereas males do so even when they are **hemizygous** for it. About half the sons of carrier (heterozygous) females are affected; in case of homozygous females showing the trait, all of her sons are likely to be affected – hence males are most affected in the population. Affected persons are related to one another through the **maternal side** of the family. Any evidence of **male-to-male** transmission of the trait rules out X-linked inheritance.

Haemophilia A (Royal disease / Bleeder's disease)

- An **X-linked recessive** disease. Protein (**factor VIII**) involved in blood clotting is affected; therefore, even a simple cut results in uncontrolled bleeding.
- **Transmission** – from an unaffected female (carrier) to male progeny. Females act as carriers of the disease, but rarely suffer from haemophilia themselves, since for a female to become haemophilic, the mother should be a carrier and the father should be haemophilic.
- X-linked recessive traits show **criss-cross / zig-zag inheritance**, i.e. the grandson inherits the trait from his grandfather through his mother, who is the carrier of the gene for the trait.
- Haemophilia is also referred to as the '**royal disease**' because several members of royal families in Europe were affected by it. An example is **Queen Victoria**.

Colour Blindness (red-green colour blindness – Daltonism)

- An **X-linked recessive** disorder due to a defect in either the red or green cone of the eye, resulting in a failure to discriminate between red and green colour; caused by mutation in certain genes present on the X chromosome.
- Occurs in about **8% of males** and only about **0.4% of females** (because females have two X chromosomes).
- The son of a woman who carries the gene has a **50% chance** of being colour blind. The mother is not herself colour blind because the gene is recessive – its effect is suppressed by her matching dominant normal gene.
- A daughter will not normally be colour blind, unless her mother is a carrier and her father is colour blind. All daughters will definitely be colour blind if both her parents are colour blind.

Y-linked trait (male sex-limited trait)

- These are traits whose gene is present on the Y-chromosome. Females do not have any Y-chromosome; all males must have a Y-chromosome, obtained from their father.
- Any trait linked to the Y-chromosome must be present only in males, and certainly not in any of the females – hence these traits are also called **male-sex limited traits**. All the sons of an affected male would express the trait, whereas none of his daughters would do so.
- Example – **Hypertrichosis** (presence of hairs on the pinna).

Chromosomal Disorders

- Chromosomal disorders are disorders characterised by changes in the **chromosome number and structure**, which cause a series of abnormalities collectively known as a **syndrome**. They may affect either the **autosomes** or the **allosomes**.
- Total number of chromosomes in humans = **46 (23 pairs)** = 22 pairs of autosomes (body chromosomes) + 1 pair of allosomes (sex chromosomes).
- **Aneuploidy** – loss or gain of chromosomes due to failure of segregation of chromatids during cell division.

- **Monosomy ($2n-1$)** – lack of one chromosome in any one pair of homologous chromosomes. **Trisomy ($2n+1$)** – inclusion of an additional copy of a chromosome in any one pair of homologous chromosomes.

Down's syndrome (Mongolism)

- First described by **Langdon Down (1866)**.
- **Cause** – presence of an additional copy of chromosome 21 (**autosomal trisomy of 21**).
- **Symptoms** – short stature, small round head, furrowed tongue, partially opened mouth, palm crease, congenital heart disease, and mental retardation.

Cri du chat syndrome ("cry of the cat")

- **Cause** – a rare syndrome caused by a **deletion of the short arm of chromosome 5** (partial monosomy) during the development of an egg or sperm.
- This syndrome derives its name from French, meaning "cry of the cat", referring to the infant's cry, which is high pitched and sounds like a cat, due to abnormal larynx development.
- **Occurrence** – a rare disease, occurring at the rate of one in 50,000 live births.
- **Symptoms** – produces a high pitched sound (cry of the cat); infants have low birth weight; may have respiratory problems; severe mental retardation; slow and incomplete development of motor skills; some people with this disorder have a shortened lifespan, but most have normal life expectancy.

Klinefelter's Syndrome

- **Cause** – an additional copy of the X chromosome, i.e. **47 chromosomes (XXY)** (allosomal trisomy).
- **Symptoms** – the individual is sterile, with overall masculine development along with **gynaecomastia** (breast development); tendency to grow fewer chest hairs; poor beard growth; wide hips; long arms and legs; narrow shoulders.

Turner's Syndrome

- **Cause** – absence of one X chromosome, i.e. **45 chromosomes (XO)** (allosomal monosomy).
- **Symptoms** – affected females are sterile; have rudimentary ovaries; secondary sexual characters are absent; short stature; low hairline; webbed neck (fold of skin); shield-shaped thorax with widely spaced nipples; constriction of the aorta; elbow deformity; rudimentary ovaries; no menstruation.